

feature-selection-stdm

August 12, 2025

1 Univariate Selecection Method @Feature Selection, Algorithm: SelectKBest Algorithm

```
[22]: import pandas as pd
import numpy as np
import seaborn as sns
```

```
[16]: #df = pd.read_csv("cardio_train.csv", sep = ';')
#df = pd.read_csv("bank.csv", sep = ';')

df = pd.read_csv("cardio_train.csv", sep = ';')
```

```
[17]: df
```

```
[17]:
```

	id	age	gender	height	weight	ap_hi	ap_lo	cholesterol	gluc	\
0	0	18393	2	168	62.0	110	80	1	1	
1	1	20228	1	156	85.0	140	90	3	1	
2	2	18857	1	165	64.0	130	70	3	1	
3	3	17623	2	169	82.0	150	100	1	1	
4	4	17474	1	156	56.0	100	60	1	1	
...	...	...	...	...	...	...	...	...	...	
69995	99993	19240	2	168	76.0	120	80	1	1	
69996	99995	22601	1	158	126.0	140	90	2	2	
69997	99996	19066	2	183	105.0	180	90	3	1	
69998	99998	22431	1	163	72.0	135	80	1	2	
69999	99999	20540	1	170	72.0	120	80	2	1	
	smoke	alco	active	cardio						
0	0	0	1	0						
1	0	0	1	1						
2	0	0	0	1						
3	0	0	1	1						
4	0	0	0	0						
...	...	...	...							
69995	1	0	1	0						
69996	0	0	1	1						
69997	0	1	0	1						

```
69998      0      0      0      1
69999      0      0      1      0
```

```
[70000 rows x 13 columns]
```

```
[19]: df.shape
```

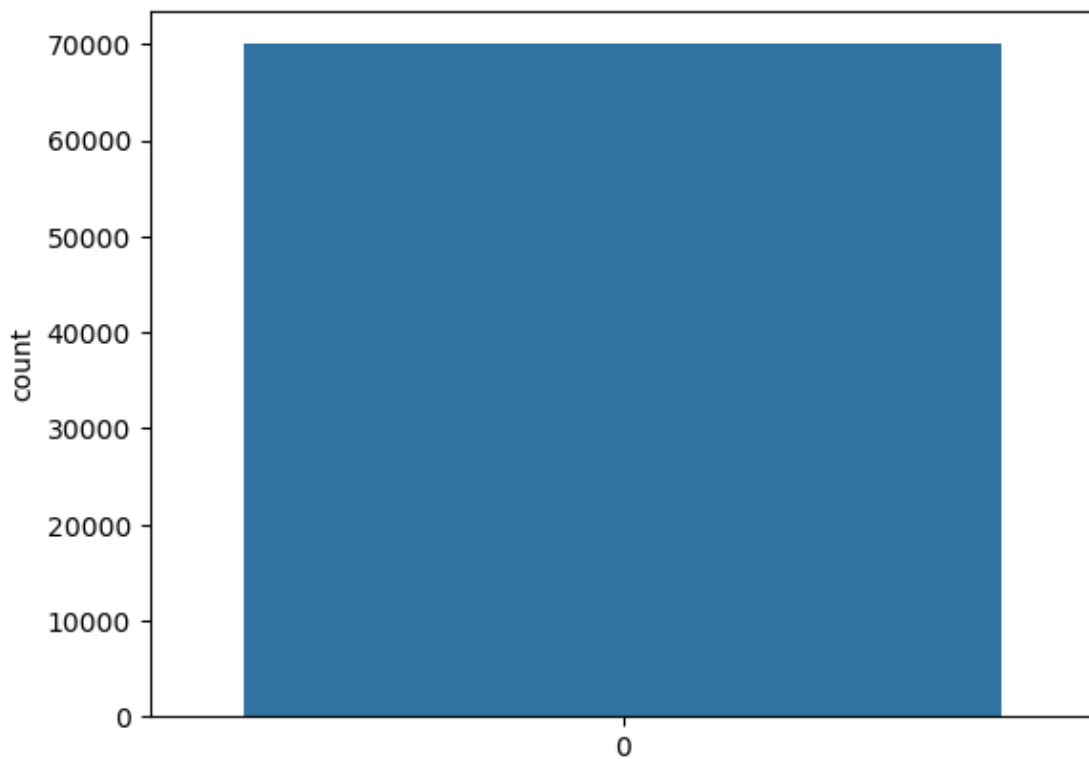
```
[19]: (70000, 13)
```

```
[20]: df["cardio"].value_counts()
```

```
[20]: cardio
0      35021
1      34979
Name: count, dtype: int64
```

```
[24]: sns.countplot(df["cardio"])
```

```
[24]: <Axes: ylabel='count'>
```



```
[25]: # x = df.iloc[:, :-1]:
#This line is selecting all rows (:) and all columns except the last one (:-1).
```

```
# : means "select all rows."
# :-1 means "select all columns except the last one." In Python slicing, :-1
↳ excludes the last item, so this will include all columns up to, but not
↳ including, the last one.
# Result: x will be a DataFrame containing all the columns of df except the
↳ last column.

x = df.iloc[:, :-1]
```

```
[27]: # y = df.iloc[:, 12]
# This line is selecting all rows (:) and the 13th column (remember, indexing
↳ in Python starts at 0, so index 12 corresponds to the 13th column).
# : means "select all rows."
# 12 means "select the 13th column" (as Python uses 0-based indexing).
# Result: y will be a Series containing the values of the 13th column of df.
```

```
Cell In[27], line 7
    y = df.iloc[:, 12]
    ~
```

```
IndentationError: unexpected indent
```

```
[28]: #y = df.iloc[:, 12]
y = df.iloc[:, 12]
```

```
[29]: from sklearn.feature_selection import SelectKBest
```

```
[30]: from sklearn.feature_selection import f_classif
```

```
[31]: FIT_FEATURES= SelectKBest(score_func = f_classif)
```

```
[32]: FIT_FEATURES.fit(x,y)
```

```
[32]: SelectKBest()
```

```
[34]: pd.DataFrame(FIT_FEATURES.scores_)
```

```
[34]:
          0
0      1.010461
1    4209.007957
2      4.603641
3      8.197397
4    2388.777887
5     208.339524
6     303.629011
```

```
7    3599.361137
8     562.772977
9      16.790541
10     3.761355
11    89.091494
```

```
[35]: score_COL = pd.DataFrame(FIT_FEATURES.scores_)
```

```
[36]: score_COL
```

```
[36]:
```

	0
0	1.010461
1	4209.007957
2	4.603641
3	8.197397
4	2388.777887
5	208.339524
6	303.629011
7	3599.361137
8	562.772977
9	16.790541
10	3.761355
11	89.091494

```
[42]: # Assuming FIT_FEATURES_scores is an array or a list of scores
score_COL = pd.DataFrame(FIT_FEATURES.scores_, columns=["scorevalue"])
# Now score_COL is a DataFrame with a single column named "scorevalue"
```

```
[43]: score_COL
```

```
[43]:
```

	scorevalue
0	1.010461
1	4209.007957
2	4.603641
3	8.197397
4	2388.777887
5	208.339524
6	303.629011
7	3599.361137
8	562.772977
9	16.790541
10	3.761355
11	89.091494

```
[46]: NEW_COL = pd.DataFrame(x.columns)
```

```
[47]: NEW_COL
```

```
[47]:
      0
0      id
1      age
2      gender
3      height
4      weight
5      ap_hi
6      ap_lo
7  cholesterol
8      gluc
9      smoke
10     alco
11     active
```

```
[52]: top_features = pd.concat([NEW_COL, score_COL], axis=1)
```

```
[53]: top_features
```

```
[53]:
      0  scorevalue
0      id      1.010461
1      age  4209.007957
2      gender    4.603641
3      height    8.197397
4      weight  2388.777887
5      ap_hi   208.339524
6      ap_lo   303.629011
7  cholesterol  3599.361137
8      gluc   562.772977
9      smoke   16.790541
10     alco    3.761355
11     active   89.091494
```

```
[55]: top_features.nlargest (8,"scorevalue")
```

```
[55]:
      0  scorevalue
1      age  4209.007957
7  cholesterol  3599.361137
4      weight  2388.777887
8      gluc   562.772977
6      ap_lo   303.629011
5      ap_hi   208.339524
11     active   89.091494
9      smoke   16.790541
```